§3 The three principles: SM, EU and FR — part 5 of 5

Reference: subsec:s3_tstab, subsec:principles_to_tipa

Source: paper/section3_principles.tex, lines 4169–4969 (§3.11 time-integrated evaluation of the three principles, including T-STAB; §3.12 connection from the three principles to structural functions and standard functional forms), at commit e6e12cd. The Japanese manuscript is authoritative. Part 5 of §3 follows 03_principles_4.md and completes §3. §4 begins in 04_tipa_1.md.

---

§3.11 Time-integrated evaluation of the three principles

Reference: subsec:s3_tstab

This subsection gives a Phase 1 formal representation that aggregates the states related to the three principles over time. It does not redefine the principles, does not change any time-indexed satisfaction predicate, and does not incorporate the relation quality Q of §6 into the predicate of EU or into the structural environment coupling condition as an additional condition.

What the paper requires structurally is that the ACS, under self–environment interaction, be sensitive to at least some of: decline of the feasibility margin, non-satisfaction of required environmental factors, degeneration of feasible responses, and feasibility-relevant differences accompanying external relations, and be able to use them in forming responses or in internal updating.

The evaluation scales, the evaluation vector s_t, the finite time window and T-STAB introduced below are conditions for description and comparison that aggregate, for a theoretical sequence of states, the states related to the three principles over time. They are neither an implementation of a function by which the ACS evaluates over time nor the output of such a function. That the ACS's internal evaluation integrates over time is a matter of the functional level; T-STAB is judged by theoretical values, not by the result of internal evaluation. The ACS is therefore not required to compute the true values of these quantities internally with the same formula, the same scale or the same precision.

Reference scale

Definition (def:s3_eval_scales): reference scale for the evaluation representation. To express SM-related evaluation normalized into $[0, 1]$, a positive feasibility-margin reference scale $\Pi_{\mathrm{ref}} > 0$, relative to the ACS concerned, is used. Π_{ref} is not a constant common to all ACSs, and identifying it with $\max_{x \in K} \Pi(x)$ is not required.

Status of the scale (rem:s3_scale_status). Π_{ref} is not a condition of any time-indexed satisfaction predicate; changing it does not change the truth of def:SM, def:EU or def:FR. It is a mathematical calibration parameter mapping differences related to the principles to degrees, and adds no component to the 7-tuple. No reference scale is placed on μ_{eff}, and μ_{eff} is not used in the conditions of T-STAB. Numerical calibration, correspondence to internal signals and scale design per kind of object are not specified in Phase 1 (within present framework).

Formal representation of the evaluation

Definition (def:s_sm): SM evaluation scale (classification C). For all $t \in \mathbb{N}_0$, $s_{\mathrm{SM},t} := \min\{1, \max\{0, \Pi(x_t)/\Pi_{\mathrm{ref}}\}\} \in [0, 1]$. If $\Pi(x_t) \leq 0$ then $s_{\mathrm{SM},t} = 0$. It expresses the degree of positive feasibility margin, not the truth of SM_t.

Definition (def:s_eu, def:s_fr): EU and FR evaluation scales (classification C). For $t \in \mathbb{T}_{\mathrm{P},i}$: $s_{\mathrm{EU},t} := 1$ if EU_t is true and 0 if false. For $t \in \mathbb{T}_{\mathrm{P},i}$ with $(x_t, e_t) \in \mathcal{O}_{\mathrm{FR}}$: $s_{\mathrm{FR},t} := 1$ if $A_t \neq \emptyset$ and 0 if $A_t = \emptyset$. At other times they are not defined.

Definition (def:S3): three-principle evaluation vector (classification C). $S_3 := [0,1]^3$. For $t \in \mathbb{T}_{\mathrm{P},i}$ with $(x_t, e_t) \in \mathcal{D}_{\mathrm{FR}}$, $s_t := (s_{\mathrm{SM},t}, s_{\mathrm{EU},t}, s_{\mathrm{FR},t}) \in S_3$. Since $s_{\mathrm{SM},t}$ is defined at all times and $s_{\mathrm{EU},t}$ on $\mathbb{T}_{\mathrm{P},i}$, all three components are defined at such times. At other times s_t is not defined.

The three components are not one physical quantity (rem:s3_noncommensurate). All three take values in $[0,1]$, but may not share one physical quantity or one measurement scale: $s_{\mathrm{SM},t}$ is a degree of positive feasibility margin, while $s_{\mathrm{EU},t}$ and $s_{\mathrm{FR},t}$ express the truth of the respective predicates. The subsection therefore neither averages the three into a single "total score" nor interprets an increase of one component as compensating a shortfall of another.

T-STAB: the three-principle time-integrated evaluation condition

Definition (def:tstab_params): T-STAB evaluation parameters. A lower reference value $\underline{s}_{\mathrm{SM}} \in (0,1]$ for the SM evaluation, a bound $\delta_{\max} \in [0,1]$ on the one-step variation, and a finite window width $W_T \in \mathbb{N}$ ($W_T \geq 1$) are used. They are reference conditions relative to the evaluation structure of the ACS concerned and do not change any time-indexed predicate.

Definition (def:tstab): T-STAB, the three-principle time-integrated evaluation condition. For $t \in \mathbb{N}_0$ let $\mathcal{W}_{T,t} := \{r \in \mathbb{N}_0 \mid \max(0, t - W_T + 1) \leq r \leq t\}$ and $\Delta\mathcal{W}_{T,t} := \{r \in \mathcal{W}_{T,t} \mid r - 1 \in \mathcal{W}_{T,t}\}$. T-STAB is judged only at times $t \in \mathbb{T}_{\mathrm{P},i}$. At such a time the ACS *satisfies the T-STAB condition* if all four of the following hold:

- **(T1) lower bound of the SM evaluation:** $s_{\mathrm{SM},r} \geq \underline{s}_{\mathrm{SM}}$ for all $r \in \mathcal{W}_{T,t}$.
- **(T2) local FR within the window:** for all $r \in \mathcal{W}_{T,t}$, $(x_r, e_r) \in \mathcal{D}_{\mathrm{FR}} \Rightarrow A_r \neq \emptyset$.
- **(T3) EU within the window:** EU_r is true for all $r \in \mathcal{W}_{T,t}$.
- **(T4) one-step variation of the SM evaluation:** $|s_{\mathrm{SM},r} - s_{\mathrm{SM},r-1}| \leq \delta_{\max}$ for all $r \in \Delta\mathcal{W}_{T,t}$. If $\Delta\mathcal{W}_{T,t} = \emptyset$ this condition holds vacuously.

For $t \in \mathbb{T}_{\mathrm{P},i}$, lem:principle_time_domain_down gives $\mathcal{W}_{T,t} \subseteq \mathbb{T}_{\mathrm{P},i}$, so EU_r in (T3) is defined at each $r \in \mathcal{W}_{T,t}$. (T1), (T2) and (T4) use only quantities defined at every time. The four conditions are therefore each defined independently.

Under (T1)–(T3) the evaluation vector ranges in $S_3^{\mathrm{T}} := \{s \in S_3 \mid s_{\mathrm{SM}} \geq \underline{s}_{\mathrm{SM}},\ s_{\mathrm{EU}} = 1,\ s_{\mathrm{FR}} = 1\}$, a product of a closed interval and a point set and a closed subset of S_3.

Proposition (prop:tstab_implies_principles): from T-STAB to the time-indexed predicates within the finite window. If the T-STAB condition is satisfied at $t \in \mathbb{T}_{\mathrm{P},i}$, then for all $r \in \mathcal{W}_{T,t}$: $r \in \mathbb{T}_{\mathrm{P},i}$, $\Pi(x_r) > 0$, and SM_r, EU_r, FR_r are all defined and true; moreover s_r is defined and $s_r \in S_3^{\mathrm{T}}$. *Proof idea:* lem:principle_time_domain_down gives $r \in \mathbb{T}_{\mathrm{P},i}$. (T1) with $\underline{s}_{\mathrm{SM}} > 0$ gives $s_{\mathrm{SM},r} > 0$, hence $\Pi(x_r) > 0$, $x_r \in K$ and SM_r true. (T3) gives EU_r true. By prop:fr_domain_eu, $(x_r, e_r) \in \mathcal{D}_{\mathrm{FR}}$, and (T2) gives $A_r \neq \emptyset$, so FR_r is true. Hence $s_{\mathrm{EU},r} = s_{\mathrm{FR},r} = 1$ and $s_r \in S_3^{\mathrm{T}}$.

Reach of the implication. The implication is limited to times inside the finite window $\mathcal{W}_{T,t}$. The paper does not state, as a general consequence of satisfaction of T-STAB, truth of the time-indexed predicates at times outside the window, or retention of feasibility over infinite time.

Proposition (prop:tstab_zero_margin): truth of the time-indexed predicates and T-STAB. If $t \in \mathbb{T}_{\mathrm{P},i}$ and SM_t, EU_t, FR_t are all true but $\Pi(x_t) = 0$, then the T-STAB condition is not satisfied at t. *Proof idea:* $\Pi(x_t) = 0$ gives $s_{\mathrm{SM},t} = 0 < \underline{s}_{\mathrm{SM}}$, and $t \in \mathcal{W}_{T,t}$, so (T1) fails. The proposition does not claim that a time with $\Pi(x_t) = 0$ exists; that truth of the predicates does not yield T-STAB is shown for times at which its antecedent holds.

The finite-window evaluation of relation quality is treated in subsec:ro_quality and is not included among the conditions of T-STAB (rem:Q_not_tstab_core).

Descriptive status of time aggregation (rem:tstab_descriptive_status). The subsection treats a condition that aggregates theoretical states related to the three principles over a finite window. The structural evaluation function — that the ACS can integrate over time the state differences related to the principles and use them in response formation or internal updating — is a separate matter at the functional level. s_t, S_3, $\mathcal{W}_{T,t}$, T-STAB and $\bar{Q}_{i \leftarrow j, W_Q, t}$

of §6 are mathematical representations fixed by theoretical values and are not an implementation or an output of an internal evaluation function. These formal quantities are therefore neither "true values the ACS knows fully" nor "goals the ACS should maximize exogenously". The subsection introduces no fourth principle, no new failure state and no stability theorem in the sense of dynamical systems. T-STAB is judged only at $t \in \mathbb{T}_{\mathrm{P},i}$, and its satisfaction gives truth of the predicates within the window (`prop:tstab_implies_principles`); the converse fails when the antecedent of `prop:tstab_zero_margin` holds.

---

§3.12 Connection from the three principles to structural functions and standard functional forms

`Reference: subsec:principles_to_tipa`

This subsection clarifies the connection at the level of description between SM, EU, FR and TIPA, PSL, ISL of §4. Its purpose is neither to redefine the principles as new mechanisms nor to map SM, EU, FR one-to-one onto M_1, M_2, M_3. Following the distinction between structural function and mathematical representation (`subsec:description_levels_global`) and the classification A/B/C (`def:abc_classification`), it gives only the descriptive connection: principle-level structural constraint → structural function → standard functional form.

Non-identity of the principle level and the mechanism level

The three principles are conceptually distinct constraint axes (`rem:principle_linkage`). The paper nevertheless places no one-to-one correspondence SM ↔ M_1, EU ↔ M_2, FR ↔ M_3. The same information integration, evaluation, memory, response formation or physical substrate can carry differences related to several principles at once; conversely, a function related to one principle may be distributed over several mechanisms, substrates and implementation sites (`rem:s3_description_levels`).

Structural functions related to the three principles

Definition (`def:principle_related_functions`): principle-related structural functions (classification A). For the ACS concerned, the roles at the functional level required to connect differences related to the principles to response formation or internal updating within self–environment interaction are called *principle-related structural functions*. At least the following seven groups are distinguished:

1. **Information acquisition and integration:** acquire available information (external environment state, internal state, memory, internal model) and integrate it into internal representations usable by later evaluation and response formation.
2. **Feasibility-sensitive evaluation:** be sensitive to at least some of the SM-related feasibility-relevant differences (decline of margin, approach to a threshold, deterioration of internal state) and convert them into evaluations usable in response formation.
3. **Evaluation and reconstruction of the possibility of response:** be sensitive to at least some of the FR-related differences (existence and degeneration of feasible responses, change of response room with internal-state differences, response differences over differences of required environment states) and convert them into evaluations usable for retaining, acquiring, recovering and reconstructing the potential for needed responses.
4. **Evaluation of environmental utilization and external coupling:** identify and evaluate at least some of the EU-related differences (satisfaction and non-satisfaction of required environmental factors and the approach to it, the possibility of responding to external environment state differences, realized external interaction, feasibility-relevant differences accompanying external relations).
5. **Response formation and selection:** using the integrated information and evaluations, construct response candidates relative to the current state and environment and connect them to effective actions or internal responses.
6. **Cross-time retention and updating of information:** retain or update past information, evaluations and response outcomes in a form usable by later evaluation and response formation, not closed at a single time.

7. **Physical sensing, acting and substrate maintenance:** acquire external and internal signals physically, connect response commands to real-world action, and maintain the physical substrate supporting operation.

These belong to the functional level of classification A in the sense of `def:abc_classification`. The enumeration does not require seven independent modules: one implementation structure may serve several functions and one function may be distributed over several structures. The principle named in a group's label indicates a main relation only and does not mean that the group relates to that principle alone (`rem:principle_function_overlap`).

Primitive functions, base functions and the seven groups (`rem:primitive_vs_principle_functions`). The primitive functions of §2 (`def:primitive_functions`: sensing, discrimination, evaluation, action) and the base functions (`def:base_functions`: functions that support the activity of the functions bearing those responsibilities and keep their bearers in existence) are re-described by the seven groups from the side of the differences related to the principles. The seven groups add no new primitive division and add no condition to CR-2 of §2. No one-to-one correspondence is placed. The main relations are: group 1 mainly concerns sensing and discrimination; groups 2–4 (principle-related evaluation) mainly concern evaluation, including the discrimination of the differences on which evaluation relies; group 5 mainly concerns action; group 6 can take part in any of the four divisions and, as persistence of the bearer of retained information, also concerns base functions; in group 7, physical acquisition of signals and connection to action concern sensing and action, and maintenance of the supporting physical substrate concerns base functions (the same process can be described with both aspects distinguished, as in §2). These relations are descriptive, and the paper does not require assigning each group to exactly one division or to a base function. As in `rem:primitive_vs_principles` of §2, this paper does not state, as a general consequence, truth of a time-indexed predicate from the operation of some function, or stoppage of a particular function from falsity of a predicate.

Functions are not the truth of the predicates (`rem:functions_not_principle_truth`). This paper does not state, as a general consequence of the existence of the structural functions of `def:principle_related_functions`, truth of SM_t, EU_t, FR_t at each time. A feasibility-sensitive evaluation function may be present while the realized state has $\Pi(x_t) < 0$; an evaluation-and-reconstruction function may be present while times with FR_t false, or with $\mu_{\mathrm{eff},t} \leq 1$, are not excluded; an environmental-utilization evaluation function may be present while times with EU_t false are not excluded. Hence existence of principle-related functions is not identified with truth of the time-indexed predicates at all times.

Related theoretical quantities versus internal computation (`rem:theory_quantity_vs_internal`). The function groups are structurally related to the differences expressed by theoretical quantities such as Π, K, A_t, $\mathrm{Crit}^{(1)}(x_t)$, $\mathcal{D}_{\mathrm{FR}}$, μ_{eff}, Q. Being related is distinct from holding or computing the true values internally, and the paper requires no function group to hold or compute them. For example, it does not require that the factors evaluated internally as needed coincide with $\mathrm{Crit}^{(1)}(x_t)$, nor that internally formed evaluations of relations coincide with Q (`rem:crit_description_level`). Mismatch between internal evaluation and the theoretical quantity (a miss or a false evaluation) is described at the mechanism level of §4.

Connection to TIPA

TIPA is a classification-B standard functional form (`rem:tipa_as_standard_form`). TIPA (Triadic Information-Processing Architecture) of §4 is a classification-B standard functional form that decomposes the structural functions of `def:principle_related_functions` and others so that they can be analyzed within the formal system. It consists of M_1, M_2, M_3, which standardly bear information integration, adaptive evaluation, and response formation and action selection. The distinctions to keep are: the information-integration function is not M_1; the feasibility-sensitive evaluation function is not M_2; the response-formation and selection function is not M_3. The left sides are requirements at the functional level; the right sides are the standard functional forms adopted by the paper.

Definition (`def:principles_tipa_connection`): functional connection from the three principles to TIPA. The standard division of roles is described as follows.

- M_1 **(information-integration mechanism).** Integrates externally and internally available signals, the internal model, memory and others, and forms internal representations usable by later evaluation. SM-related internal-state signals, EU-related information on satisfaction of required factors and external environment or object differences, and FR-related information on the possibility of response or state differences can be integration targets of M_1 as needed. The paper does not require M_1 to reconstruct the theoretical $\Pi(x_t)$, $E_{\mathrm{req}}(x_t)$, $\mathrm{Crit}^{(1)}(x_t)$, $\mu_{\mathrm{eff},t}$, Q.
- M_2 **(adaptive-evaluation mechanism).** Based on information available from M_1 and others, evaluates in a form usable for response formation feasibility threats, non-satisfaction of required factors and its approach, degeneration of response room, changes of external relations, urgency and others. Part of the differences related to any of SM, EU, FR can be an evaluation target of M_2. The paper does not require M_2 to read out directly, or compute numerically, the theoretical values Π, K, A_t, E_{req}, $\mathrm{Crit}^{(1)}$, $\mathcal{O}_{\mathrm{FR}}$, μ_{eff}, Q, s_t or T-STAB.
- M_3 **(response-action mechanism).** Using integrated information and evaluations, bears response formation: constructing response candidates, judging conformity to requirements, issuing action commands. Evaluation differences related to the principles can be reflected in candidate formation or selection by M_3 as needed. The three principles are not a single objective function given to M_3 from outside. The paper does not state, as a general consequence, truth of a time-indexed predicate from the output of M_3 alone; even if FR_t is true, the paper does not require the action command or effective action of M_3 to lie in A_t (`rem:selection_responsibility`).

M_1, M_2, M_3 are not principle-specific mechanisms (`rem:tipa_not_principle_modules`). The definition does not define M_1, M_2, M_3 as mechanisms specific to particular principles. For example, processing an SM-related danger signal can involve integration by M_1, evaluation by M_2 and response formation by M_3; likewise EU-related satisfaction of required factors and feasibility-relevant differences accompanying external relations, and FR-related degeneration of responses, can be handled across several mechanisms. The distinction among the three mechanisms is a standard decomposition by functional responsibility — input integration → evaluation → response formation — and the paper does not claim it is isomorphic to the three-way division SM, EU, FR.

Connection to PSL

Definition (`def:principles_psl_connection`): principle-related functions and PSL. PSL (Physical Substrate Layer, `subsec:psl`) is a classification-B standard functional form that provides the physical connection between TIPA and implementation sites for sensing, acting, substrate maintenance, local protective responses, and related functions. The relation to the principles is described as follows:

1. **Input side:** for external and internal signals related to SM, EU, FR to become available to the information processing of TIPA, input-side substrate functions that acquire and transmit them physically are needed.
2. **Output side:** for response commands formed in TIPA to be connected to effective action, output-side functions of physical realization are needed.
3. **Substrate maintenance:** to keep information processing, memory, sensing and acting continuing, functions maintaining the substrate that bears them are needed.
4. **Protective response to acute local anomalies:** for acute local anomalies where waiting for completion of deliberative evaluation can be inappropriate for maintaining feasibility, a local, low-latency protective response function can support principle-related operation.

The concrete standard forms are left to `subsec:psl`.

PSL functions are not the principles themselves (`rem:psl_not_principles`). No standard functional form of PSL is identical to any of SM, EU, FR. For example, although substrate maintenance relates strongly to SM, "PSL substrate maintenance $\iff$ SM" is not adopted. Although a local protective response can handle an acute feasibility threat, this paper does not state, as a general consequence of its operation alone, truth of the time-indexed predicates of the three principles including EU_t or FR_t. PSL is a standard functional form connecting principle-related functions to physical operation.

Connection to ISL

Definition (`def:principles_isl_connection`): principle-related functions and ISL. ISL (Implementation Substrate Layer, `subsec:impl`) is a classification-B descriptive layer that organizes, in the standard form of the paper, the physical implementation sites bearing sensing, acting, information retention and others. The connection to principle-related functions is described through at least three physical capabilities: sensing-capable physical carriers, acting-capable physical carriers, and information-retaining physical carriers, organized standardly in `subsec:impl` as B_{sens}, B_{eff}, B_{mem}. The paper does not require these three sets to exist in real ACSs as three mutually separate organs or independent hardware.

Implementation sites and localization of principles (`rem:principles_not_localized`). None of SM, EU, FR is taken to be localized in a single B_{sens}, B_{eff}, B_{mem} or a single physical site. Principle-related functions can hold across several processes of sensing, memory, evaluation, response and substrate maintenance. Hence the same principle-level constraint may be borne by different physical arrangements in biological, artificial or other implementations.

Division of roles with the temporal change

Responsibility boundary between §4 and §5 (`rem:s4_s5_boundary_from_s3`). This subsection connects the structural functions related to the principles to the standard functional forms TIPA, PSL, ISL. §4 treats the structure and operation of mechanisms, substrates and implementation sites at the same time. §5, without redefining the structures defined in §4, treats the deterministic change from t to $t+1$ of the internal model, functional memory, bias, effective development freedom, prediction horizon, health of PSL and ISL and others. The boundary is: §4 = structure and operation at one time; §5 = change over time of already-defined structures and internal quantities.

Update and principle-related functions (`rem:update_principle_connection`). Update of the 7-tuple expresses the update of the internal model between times. That evaluation results, memory, internal-state differences, external environment differences and others related to the principles can be reflected in the internal model at later times can be understood as a connection between principle-related structural functions and Update. The long-term aspect of FR — retaining, acquiring, recovering and reconstructing the potential for needed responses — can be described through this connection. The subsection does not specify a concrete algorithm of Update, learning rate, convergence property or optimization method; the inter-time update structure is left to §5.

Range of correspondence

Main function connections by principle can overlap (`rem:principle_function_overlap`). The main connections can be described intuitively as follows.

- **SM-related:** acquisition, integration, evaluation and response formation for signals on internal state and feasibility threats; physical protection; substrate maintenance.
- **EU-related:** acquisition and evaluation of signals on satisfaction and non-satisfaction of required factors and its approach; sensing of external environment state differences; discrimination of external input; processing of realized external interaction; evaluation of feasibility-relevant differences accompanying external relations; response formation toward satisfaction of required factors.
- **FR-related:** evaluation of the existence and degeneration of feasible responses; of changes of response room with internal-state differences; of response differences over

required-environment-state differences; memory and internal-model update; reconstruction of response candidates.

This is not an exclusive assignment table by principle. For example, sensing of external environment differences relates not only to EU but also to danger detection for SM and to maintaining response diversity for FR; memory is used not only for FR but also in evaluations related to SM and EU. The correspondence between the three principles and mechanism-level functions is therefore many-to-many.

Minimal claim of the connection from the principle level to the mechanism level (`rem: principles_to_mechanisms_scope`). What the subsection claims is a requirement at the functional level: structural differences related to the three principles must not be entirely cut off from the operation of the ACS, and at least some of them can connect to later operation through information acquisition and integration, evaluation, response formation, memory and updating, and physical sensing, acting and substrate maintenance. The paper organizes this as structural functions of classification A, and §4 describes it by the standard functional forms TIPA, PSL, ISL of classification B. The subsection thus gives a descriptive bridge: three principles → structural functions → standard functional forms. The three principles, particular mechanisms and particular physical sites belong to different levels of description and are not identified with one another. The concrete definitions of TIPA, PSL, ISL, the responsibility boundaries of input, processing and output, high-speed processing, Automaticity, physical input–output connection and the structure of implementation sites are left to §4; inter-time update, learning, health change and others are left to §5.

Check status. This summary adds no new definition, assumption or result. Labels in backticks are shared with the manuscript. It has been checked against the manuscript by the drafting model and audited by an external AI model. It remains a draft until the author checks it against the manuscript. No independent human third-party verification is claimed.